

TOPIC FOUR- SOLID WASTE AND REFUSE DISPOSAL SYSTEM

A. SOLID WASTE

Waste (also known as rubbish, trash, refuse, garbage, junk, litter) is unwanted or useless materials. In biology, waste is any of the many unwanted substances or toxins that are expelled from living organisms; such as urea, sweat or faeces. Litter is waste which has been disposed of improperly.

Waste is sometimes a subjective concept, because items that some people discard may have value to others. It is widely recognized that waste materials are a valuable resource, whilst there is debate as to how this value is best realized. Such concepts are colloquially expressed in western culture by such idioms as "One man's trash is another man's treasure."

Solid/Refuse disposal refers to all the techniques used for the collection, treatment and disposal of the solid waste of a community.

The Sustainable Waste Management Act (2022) provides the legal framework, emphasizing the "5Rs" (Reduce, Reuse, Recycle, Repair, Refurbish) and requiring counties to formalize waste management.

Classification of Solid waste

Solid wastes are the organic and inorganic waste materials such as product packaging, grass clippings, furniture, clothing, bottles, kitchen refuse, paper, appliances, paint cans, batteries, etc., produced in a society, which do not generally carry any value to the first user(s).

Solid wastes, thus, encompass both a heterogeneous mass of wastes from the urban community as well as a more homogeneous accumulation of agricultural, industrial and mineral wastes. While wastes have little or no value in one setting or to the one who wants to dispose them, the discharged wastes may gain significant value in another setting. Knowledge of the sources and types of solid wastes as well as the information on composition and the rate at which wastes are generated/ disposed is, therefore, essential for the design and operation of the functional elements associated with the management of solid wastes.

Source-based classification.

- Residential: This refers to wastes from dwellings, apartments, etc., and consists of leftover food, vegetable peels, plastic, clothes, ashes, etc.
- Commercial: This refers to wastes consisting of leftover food, glasses, metals, ashes, etc., negated from stores, restaurants, markets, hotels, motels, auto-repair shops, medical facilities, etc.
- Institutional: This mainly consists of paper, plastic, glasses, etc., generated from educational, administrative and public buildings such as schools, colleges, offices, prisons, etc.
- Municipal: This includes dust, leafy matter, building debris, treatment plant residual sludge, etc., generated from various municipal activities like construction and demolitions street cleaning, landscaping, etc.

- **Industrial:** This mainly consists of process wastes, ashes, demolition and construction wastes, hazardous wastes, etc., due to industrial activities.
- **Agricultural:** This mainly consists of spoiled food grains and vegetables, agricultural remains, litter, etc., generated from fields, orchards, vineyards, farms, etc.
- **Open areas:** this includes wastes from areas such as Streets, alleys, parks, vacant lots, playgrounds, beaches, highways, recreational areas, etc.

Type-based classification.

Classification of wastes based on types, i.e., physical, chemical, and biological characteristics of wastes, is as follows.

- **Garbage:** This refers to animal and vegetable wastes resulting from the handling, sale, storage, preparation, cooking and serving of food. Garbage comprising these wastes contains putrescible (rotting) organic matter, which produces an obnoxious odor and attracts rats and other vermin. It, therefore, requires special attention in storage, handling and disposal.
- **Ashes and residues:** These are substances remaining from the burning of wood, coal, charcoal, coke and other combustible materials for cooking and heating in houses, institutions and small industrial establishments. When produced in large quantities, as in power-generation plants and factories, these are classified as industrial wastes. Ashes consist of fine powdery residue, cinders and clinker often mixed with small pieces of metal and glass. Since ashes and residues are almost entirely inorganic, they are valuable in landfills.
- **Combustible and non-combustible wastes:** These consist of wastes generated from households, institutions, commercial activities, etc., excluding food wastes and other highly putrescible material. Typically, while combustible material consists of paper, cardboard, textile, rubber, garden trimmings, etc., non-combustible material consists of such items as glass, crockery, tin and aluminum cans, ferrous and non-ferrous material and dirt.
- **Bulky wastes:** These include large household appliances such as refrigerators, washing machines, furniture, crates, vehicle parts, tires, wood, trees and branches. Since these household wastes cannot be accommodated in normal storage containers, they require a special collection mechanism.
- **Street wastes:** These refer to wastes that are collected from streets, walkways, alleys, parks and vacant plots, and include paper, cardboard, plastics, dirt, leaves and other vegetable matter. Littering in public places is indeed a widespread and acute problem in many countries, and a solid waste management system must address this menace appropriately.
- **Biodegradable and non-biodegradable wastes:** Biodegradable wastes mainly refer to substances consisting of organic matter such as leftover food, vegetable and fruit peels, paper, textile, wood, etc., generated from various household and industrial activities. Because of the action of micro-organisms, these wastes are degraded from complex to simpler compounds. Non-biodegradable wastes consist of inorganic and recyclable materials such as plastic, glass, cans, metals, etc.
- **Construction and demolition wastes:** These are wastes generated as a result of construction, refurbishment, repair and demolition of houses, commercial buildings and

other structures. They consist mainly of earth, stones, concrete, bricks, lumber, roofing and plumbing materials, heating systems and electrical wires and parts of the general municipal waste stream.

- **Hazardous wastes:** Wastes of industrial, institutional or consumer origin that are potentially dangerous either immediately or over a period of time to human beings and the environment. This is due to their physical, chemical and biological or radioactive characteristics like ignitability, corrosivity, reactivity and toxicity. Note that in some cases, the active agents may be liquid or gaseous hazardous wastes. These are, nevertheless, classified as solid wastes as they are confined in solid containers. Typical examples of hazardous wastes are empty containers of solvents, paints and pesticides, which are frequently mixed with municipal wastes and become part of the urban waste stream. Certain hazardous wastes may cause explosions in incinerators and fires at landfill sites. Others such as pathological wastes from hospitals and radioactive wastes also require special handling. Effective management practices should ensure that hazardous wastes are stored, collected, transported and disposed of separately, preferably after suitable treatment to render them harmless.
- **Sewage wastes:** The solid by-products of sewage treatment are classified as sewage wastes. They are mostly organic and derived from the treatment of organic sludge separated from both raw and treated sewages. The inorganic fraction of raw sewage such as grit and eggshells is separated at the preliminary stage of treatment, as it may entrain putrescible organic matter with pathogens and must be buried without delay. The bulk of treated, dewatered sludge is useful as a soil conditioner but is invariably uneconomical. Solid sludge, therefore, enters the stream of municipal wastes, unless special arrangements are made for its disposal.

SOLID WASTE MANAGEMENT (SWM) SYSTEM

A SWM system refers to a combination of various functional elements associated with the management of solid wastes. The system, when put in place, facilitates the collection and disposal of solid wastes in the community at minimal costs, while preserving public health and ensuring little or minimal adverse impact on the environment. The functional elements that constitute the system are:

- **Waste generation:** Wastes are generated at the start of any process, and thereafter, at every stage as raw material are converted into goods for consumption. For example, wastes are generated from households, commercial areas, industries, institutions, street cleaning and other municipal services. The most important aspect of this part of the SWM system is the identification of waste.
- **Waste storage:** Storage is a key functional element because collection of wastes never takes place at the source or at the time of their generation. The heterogeneous wastes generated in residential areas must be removed within 8 days due to shortage of storage space and presence of biodegradable material. Onsite storage is of primary importance due to aesthetic consideration, public health and economics involved. Some of the options for storage are plastic containers, conventional dustbins (of households), used oil drums, large storage bins (for institutions and commercial areas or servicing depots), etc.





Common storage methods: Plastic bins , Sacks , Skips (construction sites)

Problem in Kenya:

- Irregular collection → overflow
- **Waste collection:** This includes gathering of wastes and hauling them to the location, where the collection vehicle is emptied, which may be a transfer station (i.e., intermediate station where wastes from smaller vehicles are transferred to larger ones and also segregated), a processing plant or a disposal site. Collection depends on the number of containers, frequency of collection, types of collection services and routes. Typically, collection is provided under various management arrangements, ranging from municipal services to franchised services, and under various forms of contracts.
- **Transfer and transport:** This functional element involve: The transfer of wastes from smaller collection vehicles, where necessary to larger ones at transfer stations. The

subsequent transport of the wastes, usually over long distances, to disposal sites. The factors that contribute to the designing of a transfer station include the type of transfer operation, capacity, equipment, accessories and environmental requirements.

Use of trucks to dumpsites . Example:Waste transported to Dandora Dumpsite

- **Processing:** Processing is required to alter the physical and chemical characteristics of wastes for energy and resource recovery and recycling. The important processing techniques include compaction, thermal volume reduction, manual separation of waste components, incineration and composting.
- **Recovery and recycling:** This includes various techniques, equipment and facilities used to improve both the efficiency of disposal system and recovery of usable material and energy. Recovery involves the separation of valuable resources from the mixed solid wastes, delivered at transfer stations or processing plants. It also involves size reduction and density separation by air classifier, magnetic device for iron and screens for glass. The selection of any recovery process is a function of economics, i.e., costs of separation versus the recovered-material products. Certain recovered materials like glass, plastics, paper, etc., can be recycled as they have economic value.
- **Waste disposal:** Disposal is the ultimate fate of all solid wastes, be they residential wastes, semi-solid wastes from municipal and industrial treatment plants, incinerator residues, composts or other substances that have no further use to the society. Thus, land use planning becomes a primary determinant in the selection, design and operation of landfill operations. A modern sanitary landfill is a method of disposing solid waste without creating a nuisance and hazard to public health. Generally, engineering principles are followed to confine the wastes to the smallest possible area, reduce them to the lowest particle volume by compaction at the site and cover them after each day's operation to reduce exposure to vermin. One of the most important functional elements of SWM, therefore, relates to the final use of the reclaimed land.

B. REFUSE DISPOSAL SYSTEMS

Some of the methods of solid/refuse waste disposal and management are as below:

- Open burning
- Dumping into the sea
- Sanitary Landfills
- Incineration
- Composting
- Ploughing in fields
- Grinding and discharging into sewers
- Salvaging
- Garchey system

For purposes of this course content, refuse chutes, incinerators, open dumps and garchey system are discussed.

1. INCINERATORS.

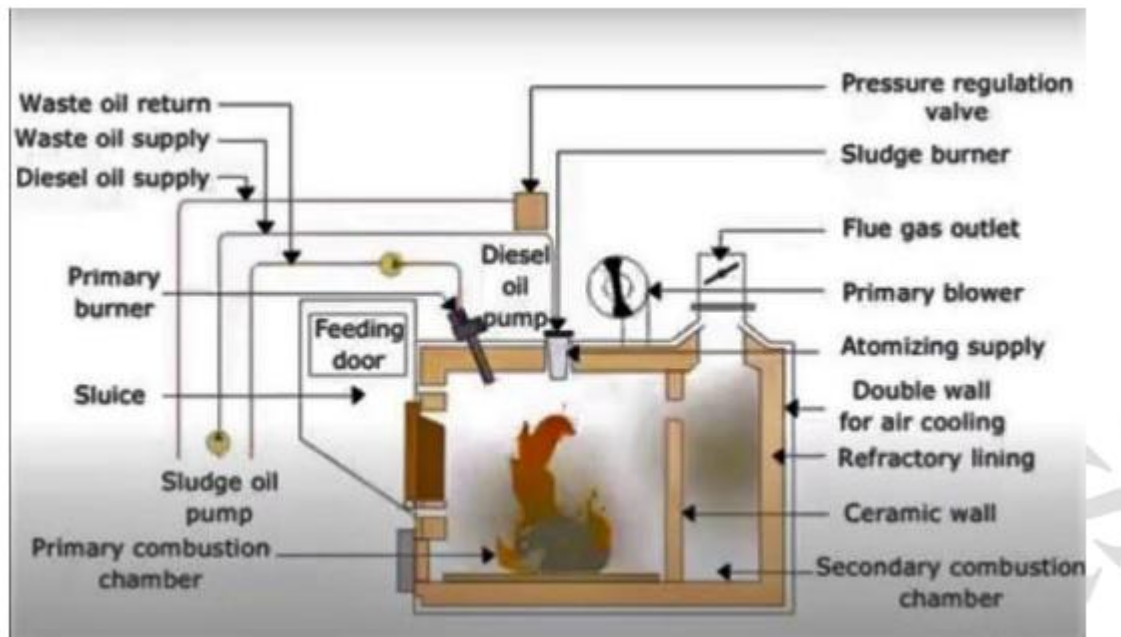
Incineration is a waste treatment process that involves the combustion of organic substances contained in waste materials. Incineration of waste materials converts the waste into ash, flue gas and heat. A large combustion chamber receives and stores the refuse until it is ignited by an automatic burner. The incineration system has a flue to discharge the incinerated gaseous products of combustion above roof level. A fan ensures negative pressure in the discharge chute to prevent smoke and fumes being misdirected.

Duration of burning is thermostatically and time controlled. Waste gases are washed and cleaned before discharging into the flue. There is no restriction on wet or dry materials, and glass, metal or plastics may be processed.

Parts of an incinerator;

An incinerator typically consists of several key components:

- a) **Primary Chamber:** This is where the waste is initially loaded and burned at high temperatures.
- b) **Secondary Chamber:** This chamber further breaks down any remaining pollutants and gases from the primary combustion.
- c) **Flue Gas Stack:** Also known as a chimney, this structure releases the gases and emissions produced during the incineration process safely into the atmosphere.
- d) **Control Systems:** These systems monitor and regulate the incineration process, ensuring optimal combustion efficiency and emissions control.
- e) **Air Pollution Control Devices:** Such as scrubbers, filters, and electrostatic precipitators, these devices remove pollutants from the flue gases before they are released into the atmosphere.
- f) **Ash Handling System:** This system collects and disposes of the ash residue left behind after incineration.
- g) **Heat Recovery System:** In some cases, heat generated during incineration is captured and used to produce steam or heat buildings, increasing the overall energy efficiency of the process.



Principle of an incinerator

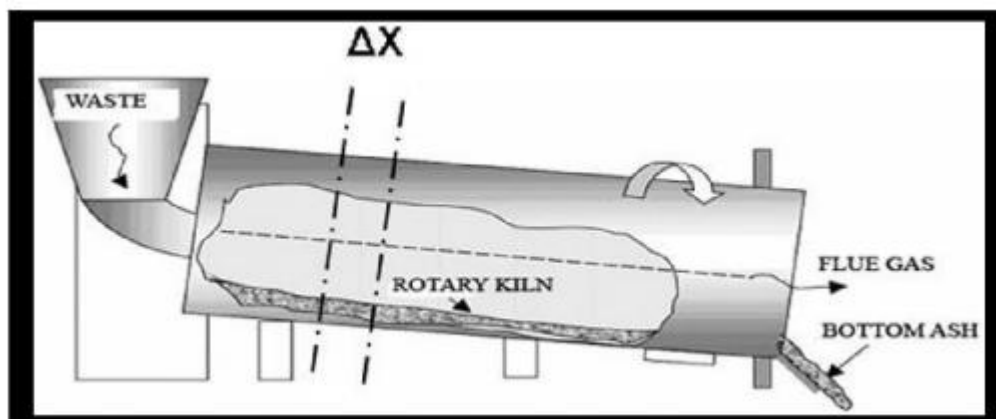
A primary combustion chamber for burning solid waste or sludge oil is included in the design of the incinerator. In contrast, a secondary combustion chamber is included for burning out any unburned exhaust gases. A primary burner is installed in the primary combustion chamber. This burner receives diesel oil to ignite it initially. After that, the primary burner receives the sludge oil until it ignites, manually or automatically turned off. The sludge burner receives atomizing air to facilitate effective combustion. On the sludge return line, there is a pressure-regulating valve that can be used to control the amount of sludge that enters the conversion

chamber. The solid waste will dry out and begin burning when the primary burner's heat is applied, and the sludge oil may also catch fire. The gases from the primary combustion chamber will burn out in the secondary combustion chamber. The primary and secondary combustion chambers are divided by a wall of ceramic heavy-duty refractory liners. Solid waste is fed into the incinerator through the feeding door in cases of solid waste or garbage. The primary burner cannot be lit if this door is open, but the rest of the combustion process continues as usual.

Types of incinerators

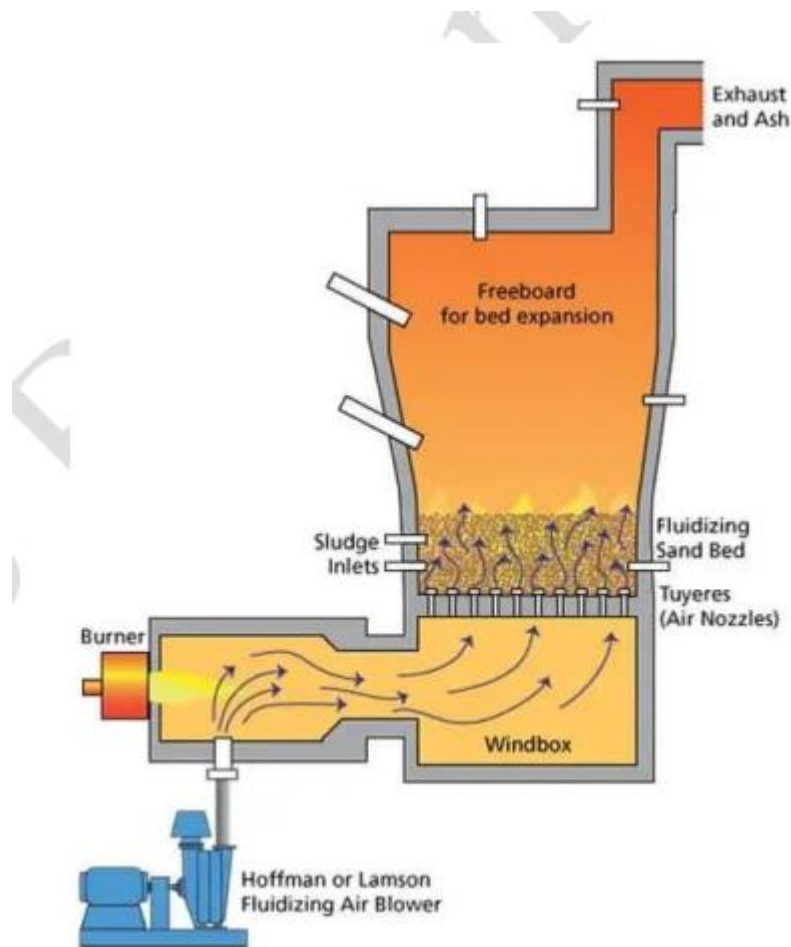
a) Rotary Kiln Incinerator

The basic design consists of two thermal treatment chambers: a primary chamber that is slightly slanted and into which waste is fed (along with hot exhaust air containing oxygen), rotated, and thermally decomposed by heat radiation from the secondary chamber; and a re-combustion chamber that is located at the back of the kiln and into which the decomposition air and the remaining waste is completely burned with the supply of secondary air. It is built with a rotating combustion chamber that keeps the garbage circulating continually. The waste can vaporize, which makes burning easier. By modifying the inclination angle and rotational rate of the kiln, the waste throughput rate can be managed. Solids burnout rates and particulate entrapment in the flue gas are higher for rotary kiln incinerators than for other incinerator designs due to the waste's turbulent motion in the primary chambers. Because of this, gas cleaning accessories are frequently added to rotary kiln incinerators.



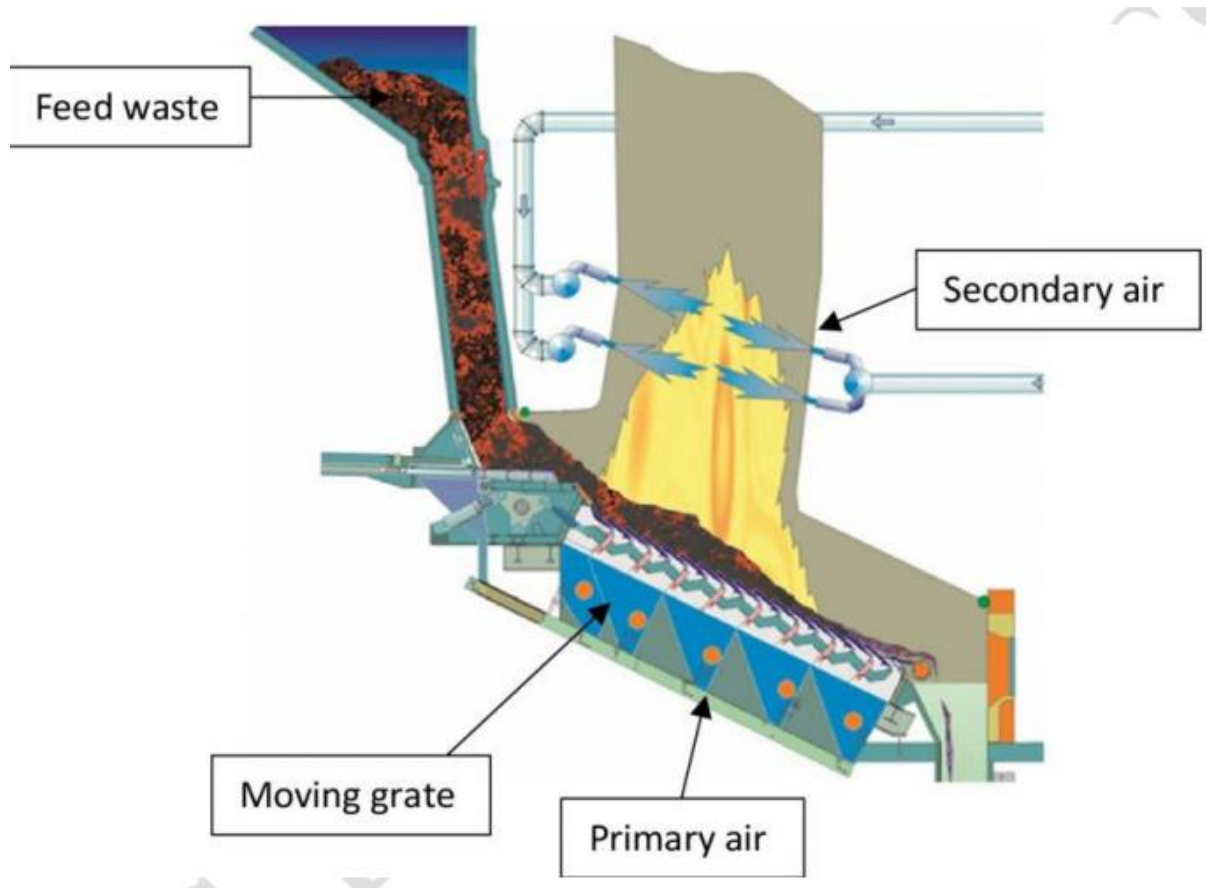
In a rotary kiln incinerator, typical waste types that can be treated include pharmaceutical waste, Hazardous waste, and medical waste, such as sharps, infectious waste, surgical trash, clinical waste, laboratory waste, etc. The advantages of rotary kilns include less Nitrogen oxides (NO_x) production and heat elimination of dangerous compounds.

b) Fluidized Bed Incinerator



The process by which a fluid is passed through granular material in a solid state to change it into a fluid-like state is known as the principle of fluidization. The reactor typically comprises a vertical steel vessel lined with refractory material and a bed of granular material, such as limestone, silica sand, or ceramic material. Air is injected into the bed through several nozzles at the bottom of the incinerator to keep it fluid. This results in a violent agitation of the bed material where the burning of trash takes place in close contact with the bed material and combustion air. Because of the comparatively low extra air level made possible by this, a high thermal efficiency of up to 90% is possible. These are employed in the incineration of sewage sludge and industrial waste and are primarily used for homogenous waste types, including liquid waste.

c) Moving Grate Incinerator



The waste is burned in layers on the grate that moves the waste through the furnace in a traditional mass-burning incinerator based on a moving grate. On the grate, the waste is dried before being burned at a high temperature with an air supply.

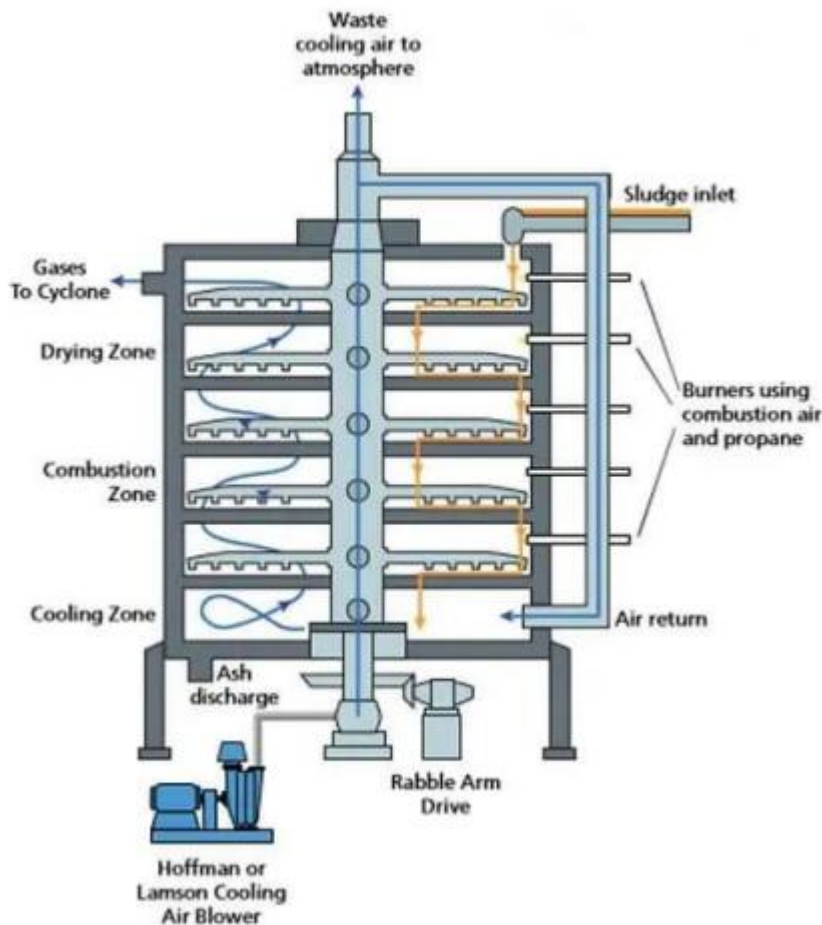
The non-combustible waste fractions, which include the ash, exit the grate through the ash chute as slag (bottom ash). The bottom ash (slag), which drops into the water trap of the slag pusher from the grate's end, is then cooled by coming into contact with cooling water before moving on

to the conveyor system. Typically, 10 to 25 percent of the weight of the waste feed is made up of slag. The main benefits of the moving grate are its well-proven technology, ability to accept wide variations in waste composition and heat values, and ability to be produced in extremely large units. The primary drawback is the expense, which is relatively significant for investment and maintenance.

d) Waste-Gas Flare Incinerator

In many industrial activities, including oil-gas extraction, refineries, chemical plants, the coal industry, and landfills, undesirable or excess gases and liquids are discharged. Gas flaring is a combustion device used to burn these gases and liquids. A substantial source of greenhouse gas emissions is gas flaring. Additionally, it produces heat and noise and renders a lot of space uninhabitable. It is used for non-hazardous waste with high organic content.

e) Multiple Hearth Incinerator

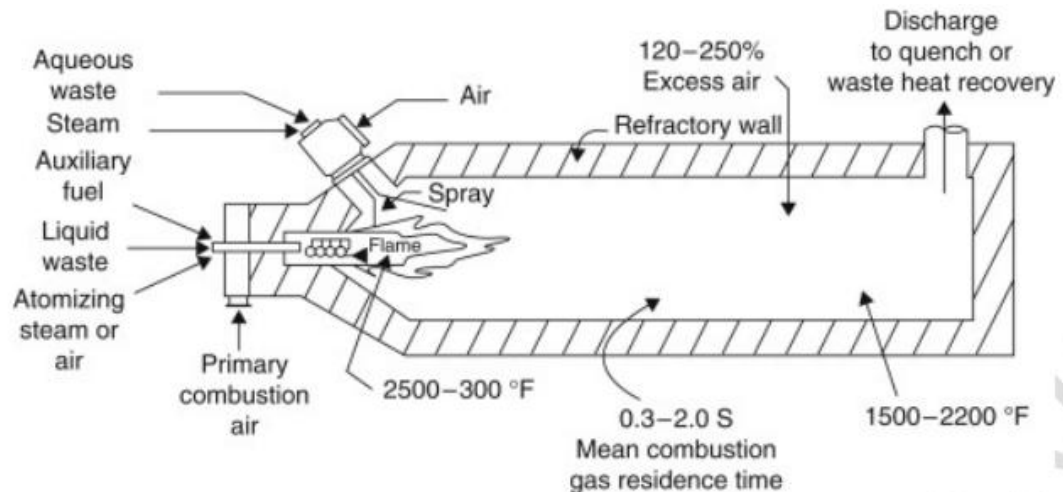


A multiple hearth incinerator consists of a row of circular hearths arranged in a column and protected by a steel shell lined with refractory with operating temperatures ranging from 1400 to 1800°F. The feed material is moved across each hearth in a spiral pattern by rabble arms attached to a vertical rotating shaft that runs through the furnace's center before dropping through the openings into the hearth below. The finished product is ejected through an outlet after traversing through each hearth.

f) Catalytic combustion chamber

In the 1970s, Engelhard Corp. discovered the procedure. It is a chemical process that uses a catalyst to accelerate the desired oxidation reactions of fuel and limit the creation of undesirable products, especially polluting nitrogen oxide emissions (NO_x), well below what is possible without catalysts. It is intended specifically for low organic concentration waste

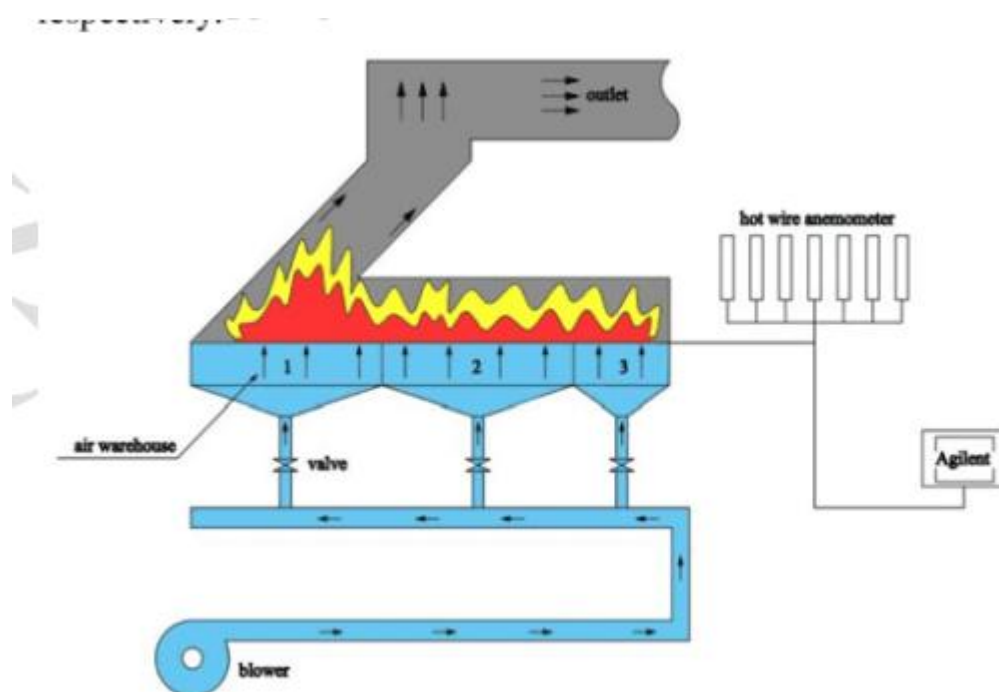
h) Liquid Injection Incinerator



A frequently used device that uses high pressure to split liquid wastes into tiny droplets for rapid burning. Wastes are delivered through nozzles and atomized into small droplets to enable the greatest possible mixing with air with operating temperatures ranging from 1200 to 3000°F. The feed must behave as a liquid with a viscosity of no more than 10,000 ssu. A solid can be used if it can be melted and pumped and can completely combust organic combustible material and non-combustibles like impurities in water.

h) Fixed Grate/ Direct-Flame Incinerator

It is often the most widely produced and utilized. Fuel and oxygen are used to maintain the direct flame, allowing trash to be burned. When waste gas contains particles, direct flame incinerators are employed. They run between 1000 and 1500 °F.



Applications of incinerator

- Medical waste; generated from laboratories, hospitals, and health clinics can pose health risks if not properly treated or disposed of. Incinerators are designed to destroy and neutralize medical waste.
- Animal waste; incinerators are employed to dispose of animal wastes, including carcasses, waste, and by-products, thereby preventing the spread of infection and viruses.
- General waste; include less harmful wastes from sites such as mining, small communities, and military operations, which must be destroyed to reduce environmental impact. Thus, incinerators are devised to incinerate such wastes.
- Nanomaterial wastes; incinerators are utilized to dispose of nanomaterial wastes such as consumer products in municipal solid waste, wastes from nanotechnology research and development, hazardous wastes, etc.

Merits of incinerator

- Incinerators can reduce the total amount of garbage by up to 95% and the solid portion of the original waste by up to 80–85%.
- Can be used to dispose of difficult-to-recycle waste
- Waste management can be facilitated and improved upon significantly through incineration. Landfills only permit organic breakdown, while non-organic waste continues to build up.
- Waste is converted into energy in incinerators, which can be utilized to produce heat or power. Can generate electricity from waste
- Compared to landfills, solid waste incinerators are less likely to contaminate the environment. Waste incineration facilities can be located close to cities or towns because they require less space. This has the advantage of reducing the need for long drives to dump waste.
- Waste is burned inside a facility where the byproducts of the incineration process can be regulated. Therefore, waste incineration plants generate a less unpleasant odor than landfills. This is because open-air degradation of waste, which adds to air pollution, is avoided.
- Waste incineration facilities are safer and better for the environment because they don't emit methane.
- Incineration facilities burn processed waste at extremely high temperatures, killing hazardous bacteria and chemicals. Can destroy harmful pathogens and toxins
- Since incinerators burn waste without leaks, they are not restricted by variations in the weather.

Demerits of incinerator

- Can produce air pollution, including nitrogen oxides, sulfur oxides, and particulate matter.
- Can produce bottom ash and fly ash, which may contain harmful metals and other pollutants.
- Not a sustainable waste management solution, as it does not conserve resources

- An incinerator plant's construction requires expensive infrastructure and equipment, which drives up the cost of the installation procedure. Can be expensive to build and operate.
- Even though garbage incinerators emit far fewer pollutants than landfills do, they nonetheless have a considerable impact on pollution, especially in the vicinity.
- Communities with waste incinerators are more likely to experience long-term negative health effects like cancer, birth defects, reproductive dysfunction, neurological issues, and other conditions known to occur at very low exposures to many of the metals and pollutants released by incinerators.
- Operation and administration of incinerators demand expensive and highly experienced labor.

2. REFUSE CHUTES

A refuse chute is an inclined channel in which refuse can be passed down from the opening of each floor to the central refuse room on the ground floor of a building.

The most common type is the dry-garbage chute which consist of a shaft with a ventilation pipe at the top, leading hatches and receiving area with containers or cans

The shaft (made from asbestos/cement pipes) which are 400mm in diameter should be vertical and have a smooth inner surface and sound insulating casing

The loading hatches/feeders of the chutes are located in stairwells or apartment kitchen. The receiving area is located beneath the chute on the ground floor or in the basement of the building

It should be at least 2.5m high, 1.5m wide and 2.5m long to provide space for garbage containers.

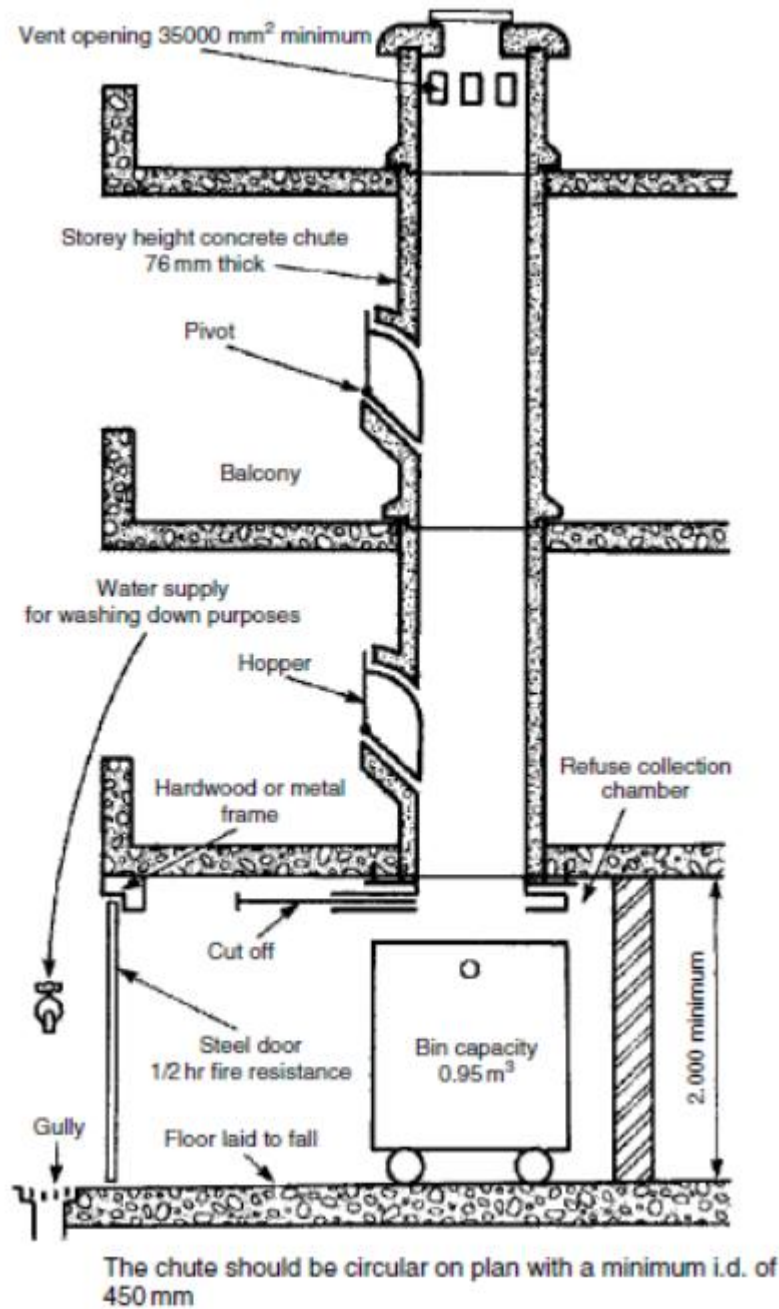
The quantity and location of refuse chutes depends upon:

- layout of the building
- number of dwellings served – maximum being six per hopper
- type of material stored
- frequency of collection
- volume of refuse
- Refuse vehicle access -within 25 meters

The chute should be sited away from habitable rooms, but not more than 30 m horizontal distance from each dwelling.

Chute linings are prefabricated from refractory or Portland cement concrete with a smooth and impervious internal surface.

The structure containing the chute void should have a fire resistance of 1 hour. The refuse chamber should also have a 1-hour fire resistance and be constructed with a dense impervious surface for ease of cleaning.



Parts of a refuse chute system

Chutes

This is a vertical pipe system passing from floor to floor provided with ventilation and inlet openings, for receiving refuse from successive flats and ending at the ground floor on the top of the collecting chamber.

Chutes may be of masonry, R.C.C. pipe, asbestos cement pipes or suitable non-corrosive material, preferably stainless steel. It should be smooth internal finish, high ductility and alkali resistant properties.

The vent shall be installed at roof level to remove foul smell and gases generated.

Inlet hopper

A receptacle fitting for receiving refuse from each flat and dropping it into the chute.

Collection chamber

This is a compartment situated at the lower end of the chute for collecting and housing the refuse during the period between two successive clearings.

The floor of the chamber shall be provided with drainage through a 100mm diameter trap and screen to prevent any solid matters flowing into the drain and the drain shall be connected to the sewer line.

It shall be provided with appropriate shutter to prevent access of scavenging animals like rats.

Types of refuse chutes in use include:

- **Vertical Chutes:** These are straight chutes that run vertically from upper floors to lower floors or a central collection point. They are the most common type and are often used in high-rise buildings.
- **Angled Chutes:** Also known as inclined chutes, these are angled at a specific degree to facilitate the smooth flow of waste from one level to another. They are useful when vertical installation is not feasible due to space constraints or building design.
- **Curved Chutes:** These chutes have a curved design, allowing waste to navigate around corners or obstacles within the building. They are commonly used in buildings with complex layouts or where straight chutes are not practical.
- **Recycling Chutes:** Specifically designed for recycling purposes, these chutes segregate different types of recyclable materials such as paper, plastic, glass, and metal. They help streamline the recycling process and promote sustainability.
- **Compactor Chutes:** These chutes are integrated with waste compactors, which compress the waste before it reaches the collection point. Compactor chutes are ideal for buildings with limited space for waste storage and can help reduce the frequency of waste removal.
- **Odour-Control Chutes:** Equipped with odor-control mechanisms such as seals, filters, or deodorizers, these chutes minimize unpleasant odors emanating from the waste. They are particularly beneficial in residential buildings where odor management is essential for occupant comfort.
- **Fire-Rated Chutes:** Designed to meet fire safety regulations, these chutes are constructed using fire-resistant materials and feature fire-rated doors or closures to prevent the spread of flames in case of a fire outbreak. They are essential for maintaining building safety standards.

Advantages of refuse chutes:

- **Convenience:** Refuse chutes provide a convenient way for occupants to dispose of waste without having to carry it to a central collection point. This can save time and effort, especially in multi-story buildings.

- **Space Efficiency:** By centralizing waste disposal, refuse chutes help optimize space within buildings, particularly in high-rise structures where storage space may be limited.
- **Improved Hygiene:** Refuse chutes help maintain cleanliness within buildings by containing waste in enclosed spaces, reducing the risk of spills, odours, and pest infestations.
- **Safety:** Carrying heavy or bulky waste items down stairwells or elevators can pose safety hazards. Refuse chutes eliminate the need for manual transportation of waste, reducing the risk of accidents and injuries.
- **Aesthetics:** Concealed within building structures, refuse chutes are often designed to blend seamlessly with interior décor, maintaining aesthetic appeal.

Disadvantages of refuse chutes:

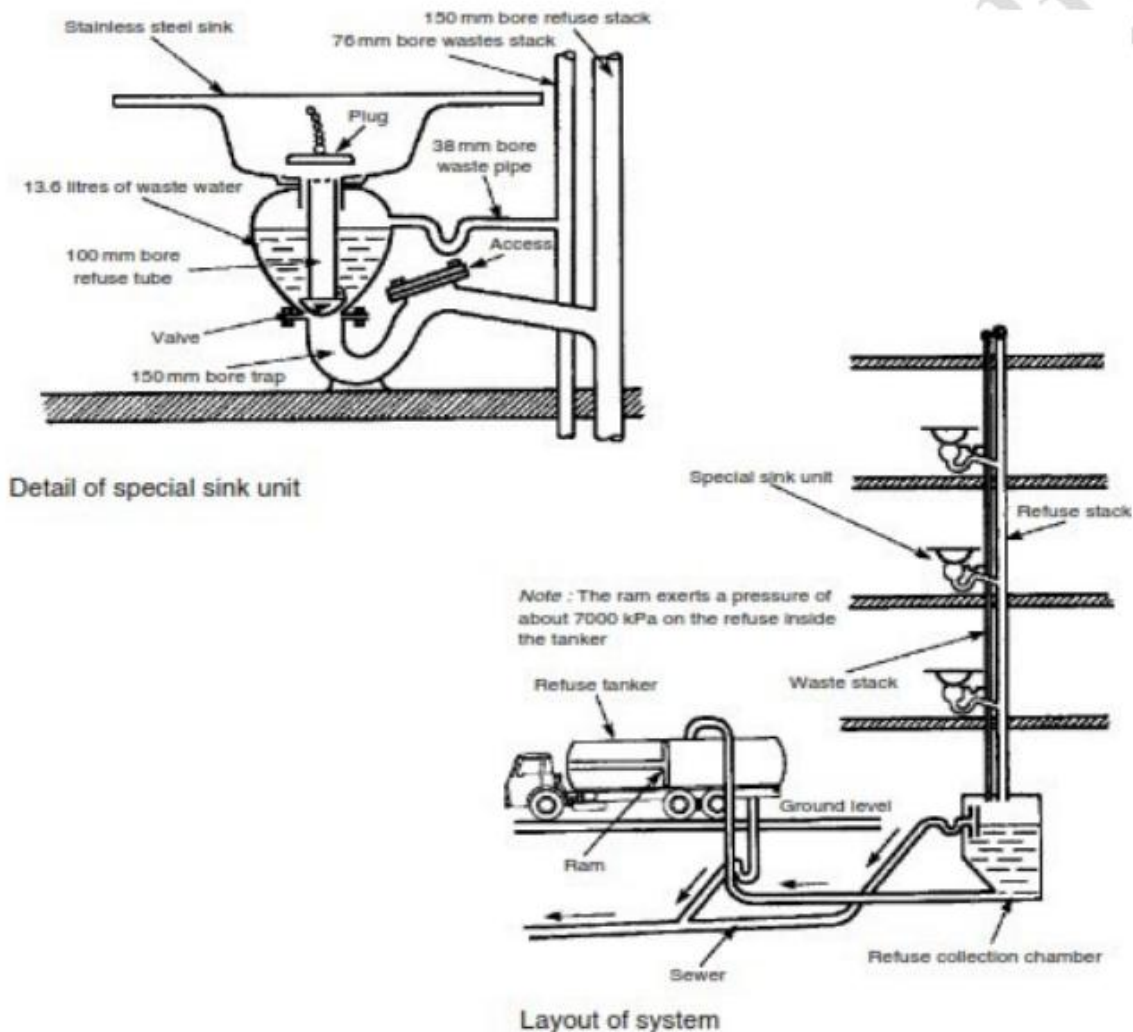
- **Odour Accumulation:** Despite efforts to control odours, refuse chutes can sometimes become sources of unpleasant smells, especially if waste is not disposed of properly or if the chute is not adequately maintained.
- **Blockages:** Refuse chutes may become clogged or blocked due to improper disposal of items, leading to operational disruptions and the need for maintenance or cleaning.
- **Cross-Contamination:** In buildings with single chutes for mixed waste, there's a risk of cross-contamination between different types of waste, potentially compromising recycling efforts or leading to environmental pollution.
- **Fire Hazards:** Accumulated waste in chutes can pose a fire hazard, especially if combustible materials are improperly disposed of or if the chute lacks fire safety features.
- **Limited Capacity:** Refuse chutes may have limited capacity, especially during peak usage periods or in buildings with high waste generation rates. This can result in overflow or delays in waste disposal.

3. GARCHEY SYSTEM

In the Garchey system, refuse (ranging from potato peelings and ashes to small bottles and tins) is collected in a unit below the sink in each residential unit through a large plug in the sink.

A bowl beneath the sink retains the normal waste water. Refuse is placed inside a central tube in the sink. When the tube is raised the waste water and the refuse are carried away down a stack or discharge pipe to a chamber at the base of the building.

Refuse from the chamber is collected at weekly intervals by a specially equipped tanker in which the refuse is compacted into a damp, semi-solid mass that is easy to tip. One tanker has sufficient capacity to contain the refuse from up to 200 dwellings. Waste water from the tanker is discharged into a foul water sewer.



Merits of Garchey system

- The standard of hygiene in terms of vermin control has reportedly been significantly higher compared to other method of waste disposal.
- The infrastructure pipework also serves for the disposal of waste water and surface run-off
- The Garchey system relies on a sealed system hence reducing the smell and vermin problems.
- Its ability to transport waste material further away from the source helped with separating the refuse collection points from the immediate vicinity of residential blocks.
- It provides a clean and efficient method of disposing of kitchen waste including also tins and bottles of a certain size.
- **Hygiene:** Enclosed system minimized odour and vermin issues.
- **Convenience:** Residents avoided carrying waste bags, especially in high-rise buildings.

- **Space-saving:** No need for individual bins or large bin rooms.
- **Potential for recycling:** Centralized collection facilitated sorting and treatment.

Demerits of Garchey system

- The dry chute method of waste disposal was associated with unhygienic and foul-smelling hoppers with equally undesirable communal hopper rooms and refuse collection areas
- Garchey evacuation tank housed below the kitchen sink effectively takes up a whole kitchen cabinet thereby reducing the storage capacity of the compact kitchens.
- Difficulty of operation. For example, blockage of the system and back flowing in lower units of buildings discourages client from using the system.
- There is also difficulty in separation of refuse types for recycling purposes
- Maintenance: The complex pipe network prone to blockages and costly repairs.
- Limited waste type: Unsuitable for bulky items, hazardous materials, or dry waste.
- Odor control: While better than open chutes unpleasant, smells could arise if not properly maintained
- Energy consumption: Water usage for flushing added to operational costs.
- Modern alternatives: More efficient and flexible waste management systems emerged.

4. LANDFILLS

Once solid waste is transported off-site it is normally taken to a landfill site. Here the waste is placed in a large excavation (pit or trench) in the ground, which is backfilled with excavated soil each day waste is tipped. Ideally, about 0.5m of soil should cover the deposited refuse at the end of the day to prevent animals from digging up the waste and flies from breeding.

The ultimate goal for land disposal should be to separate disposal of hazardous and non-hazardous materials and the construction of clean and properly sited landfills with diligent management. When these conditions are met, the landfill becomes a sanitary landfill.

The following factors are key in the establishment of sanitary landfills:

- **Siting.** Siting is possibly the most difficult stage in landfill development. Carry out an environmental impact assessment that addresses all siting criteria and organize full community involvement.
- **Design.** To mitigate environmental impacts, sanitary landfill designs should include an impermeable lining, gas monitoring, extraction, and treatment, fencing to control access and provisions for closure and post-closure monitoring and maintenance.

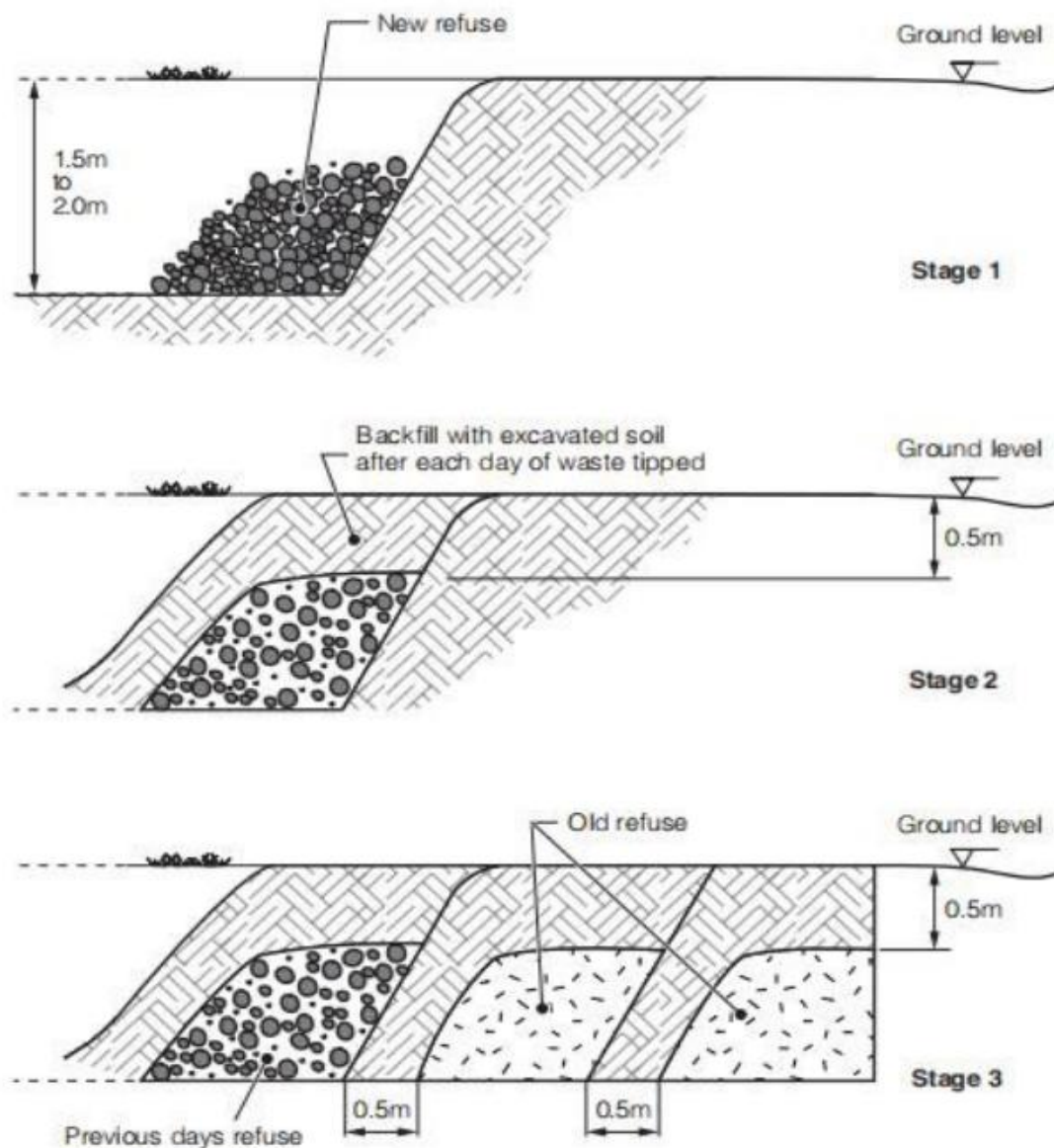
Constructing Landfills

The basic element of a sanitary landfill is the refuse cell. This is a confined portion of the site in which refuse is spread and compacted in thin layers. Several layers may be compacted on top of one another to a maximum depth of about 3 meters (10 feet). The compacted refuse occupies about one-quarter of its original loose volume.

At the end of each day's operation, the refuse is covered with a layer of soil to eliminate windblown litter, odour, and insect or rodent problems. One refuse cell thus contains the daily volume of compacted refuse and soil cover. Several adjacent refuse cells make up a lift, and

eventually a landfill may comprise two or more lifts stacked one on top of the other. The final cap for a completed landfill may also be covered with a layer of topsoil that can support vegetative growth.

Daily cover soil may be available on-site, or it may be hauled in and stockpiled from off-site sources. Various types of heavy machinery, such as crawler tractors or rubber-tired dozers, are used to spread and compact the refuse and soil. The area and depth of a new landfill are carefully staked out, and the base is prepared for construction of any required liner and leachate-collection system.



Advantages of Landfills

- **Convenience:** Landfills are a relatively simple and well-established technology for waste disposal. They require less specialized equipment and expertise compared to other methods like incineration or composting.
- **Cost-effectiveness:** Landfilling can be cheaper than other disposal methods, especially for mixed waste streams that are difficult to recycle or compost. Transportation costs can also be lower if the landfill is located nearby.
- **Sanitation:** Landfills can help to keep our communities clean and free of litter by centralizing waste disposal. Properly managed landfills can prevent the spread of disease and protect public health.
- **Energy production:** Landfill gas, produced by decomposing organic matter, can be captured and used to generate electricity or heat. This can be a valuable source of renewable energy.
- **Land reclamation:** Once a landfill is full, it can be capped and converted into usable land, such as parks or green spaces.

Disadvantages of Landfills

- **Environmental impact:** Landfills can have a significant negative impact on the environment. They can contaminate soil and groundwater with leachate, a toxic liquid that seeps from decomposing waste. Landfill gas can also contribute to greenhouse gas emissions and air pollution.
- **Limited lifespan:** Landfills have a limited lifespan and eventually fill up. Finding new suitable locations for landfills can be challenging, especially in densely populated areas.
- **Not a sustainable solution:** Landfilling does not conserve resources and buries valuable materials that could be recycled or composted. It is generally seen as a less sustainable solution compared to waste reduction, reuse, and recycling.
- **Health risks:** Living near landfills can pose health risks due to exposure to air and water pollution. Additionally, improper management of landfills can attract vectors like rodents and insects, further impacting public health.
- **Social and aesthetic concerns:** Landfills can be unsightly and create unpleasant odors, negatively impacting nearby communities. They can also displace people and communities.